22CH203 ENGINEERING CHEMISTRY-II

UNIT I

LESSON PLAN 2-Topic-1

PRIMARY REFERENCE ELECTRODE

ELECTROCHEMISTRY

Prediction of cell potentials: Standard electrodes - Determination of single electrode potential

REFERENCE ELECTRODES (OR) STANDARD ELECTRODES

A reference electrode is an electrode that maintains a stable and known electrochemical potential against which the potential of another electrode can be measured. It provides a fixed reference point for the measurement of potential in an electrochemical cell. In various electrochemical applications such as pH measurement, potentiometric, and corrosion studies, reference electrodes play a critical role.

Types of electrodes

1. Primary reference electrode. Eg. Standard Hydrogen electrode (SHE).
2. Secondary reference electrode. Eg. Calomel electrode.
3. Glass electrode (or) Indicator electrode (or) Ion-selective electrode.

1. Standard Hydrogen Electrode (Primary Reference Electrode)

The standard hydrogen electrode is the standard measurement of electrode potential for the thermodynamic scale of redox potentials. The standard hydrogen electrode is often abbreviated as SHE or may be known as a normal hydrogen electrode (NHE). Technically, a SHE and NHE are different. The NHE measures the potential of a platinum electrode in a 1 M acid solution, while the SHE measures the potential of a platinum electrode in an ideal solution (current standard of zero potential at all temperatures).

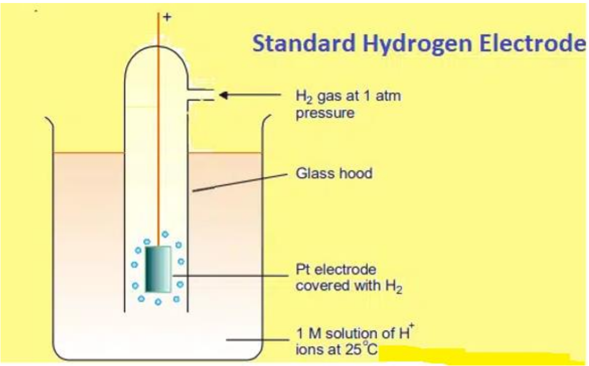

Figure 1 Standard Hydrogen Electrode

The redox reaction takes place at the platinized platinum electrode. When the electrode is dipped into the acidic solution, gaseous hydrogen at a pressure of 1 atm is bubbled over the platinum electrode, which is coated with a very finely divided platinum that causes the electrode reaction to occur rapidly. The electrode is surrounded by a solution whose temperature is 25-degree Celsius and in which the H+ ion concentration is 1M.

The half-cell reaction that occurs at the platinum surface is shown below:

Half-cell reaction of SHE

$$2H^{+} (aq, 1M) + 2e^{-} \rightleftharpoons H_2 (g, 1atm)$$

Construction

A standard hydrogen electrode (SHE) has four components:

1. Hydrogen Gas/Blow
2. Platinized platinum electrode
3. Acid solution with molarity equal to 1 mol dm^{-3} (activity H+ = 1 mol dm^{-3})
4. Reservoir to attach the second half-element of the galvanic cell

The choice of platinum for the hydrogen electrode is based on a number of factors:

- Platinum is an inert catalyst. So, it doesn't corrode during the electrochemical and speeds up the rate of the reaction.
- Platinum is a less reactive metal. So, it doesn't easily react with other metals. As a result, it provides the surface for the redox reaction.
- Great reproducibility of the potential (bias of less than 10V when two well-made hydrogen electrodes are compared)
- High intrinsic exchange current density for proton reduction on platinum
- Increase total surface area, the surface of platinum is platinized (i.e., covered with a layer of fine powdered platinum, also known as platinum black). This enhances reaction kinetics and allows for the most current possible.
- Other metals, such as palladium and hydrogen, can be used to make electrodes with comparable functions.
- Unlike other metals such as gold, silver, copper, mercury; platinum doesn't poison the electrode of another half-cell.

Working of a SHE

The standard electrode potential of any other electrode is obtained by combining the electrode with SHE. Since the **standard potential of the hydrogen electrode is zero**, the measured potential is the standard potential of the other electrode.

A standard hydrogen electrode (SHE) can function as both an anode and a cathode.

If we constructed a galvanic cell using SHE where **SHE acts as cathode** then, the cell notation can be expressed as:

Cell notation when SHE is used as cathode

$$M \,|\, M^{n+}\ (aq.) \,||\, H^{+}(aq,\ 1M) \,|\, H_2\ (g, 1atm),\ Pt$$

Here, **oxidation** takes at M/Mn^{+} electrode, and **reduction** takes place at SHE.

At metal electrode: $M_{(s)} \rightleftharpoons M^{n+} + ne^{-}$

At SHE: $nH^{+} + ne^{-} \rightleftharpoons \frac{n}{2} H_2\ (g)$

Overall reaction: $M_{(s)} + nH^{+} \rightleftharpoons \frac{n}{2} H_2\ (g)$

Similarly, if a galvanic cell is constructed using **SHE where it acts as an anode** then, the cell notation can be expressed as:

Cell notation when SHE is used as Anode

$$Pt, H_2\ (g, 1atm) \,|\, H^{+}(aq,\ 1M) \,||\, M^{n+}\ (aq.) \,|\, M$$

Here, **oxidation** takes place at SHE, and **reduction** takes place at M/Mn^{+} electrode.

At SHE: $\frac{n}{2} H_2\ (g) \rightleftharpoons nH^{+} + ne^{-}$

At metal electrode: $M^{n+} + ne^{-} \rightleftharpoons M_{(s)}$

Overall reaction: $\frac{n}{2} H_2\ (g) \rightleftharpoons M_{(s)} + nH^{+}$

Application of Standard hydrogen electrode

SHE can be used in the determination of standard electrode potential. By convention, the standard electrode potential is always taken as the standard reduction potential.

1. Determination of the standard electrode potential of the Zn^{++}/Zn electrode.

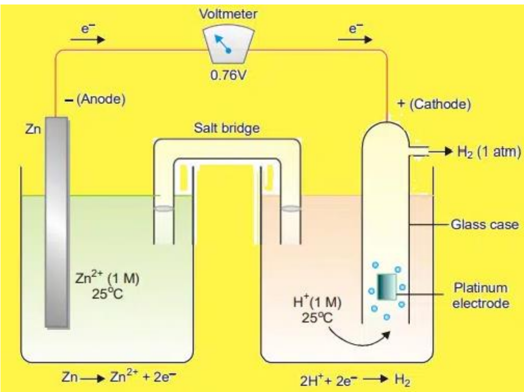

Figure.2 Standard electrode potential of the Zn^{++}/Zn electrode

In this case, electrons flow from Zn-electrode to the hydrogen electrode. Therefore, Zn-electrode acts as an anode and SHE acts as a cathode. The cell may be represented as,

$$\mathbf{Zn \mid Zn^{2+}\,(1M) \parallel H^{+}(1M) \mid H_2\,(g,1atm),\,Pt}$$

The emf of the cell is found to be 0.76V. Now, the emf of a cell is given by the following expression:

$E^o_{cell} = E^o_{Right} - E^o_{Left}$

$0.76 = E^o_{H^+/H_2} - E^o_{Zn^{++}/Zn}$

$0.76 = 0 - E^o_{Zn^{++}/Zn}$

$\mathbf{E^o_{Zn^{++}/Zn} = -0.76\ V}$

Thus, by combining with SHE, the standard electrode potential of the Zn^{++}/Zn electrode is found to be -0.76 V.

Advantages of SHE

- Due to minute potential, the potential of a standard hydrogen electrode is taken as zero, therefore, acts as reference point for other metals.
- SHE performs the dual function as a cathode half-cell as well as anode half-cell.
- It can be used to determine the unknown potential of the half cell.

Disadvantages of SHE

- Difficulty in transportation, construction, and maintenance.
- Difficulty in maintaining the pressure of hydrogen gas.
- Difficulty in levelling the concentration of the acid solution.
- Difficulty in obtaining pure hydrogen gas.
- Difficulty in the construction of an ideal platinum electrode.
- Impurities reduce the life cycle of platinum electrode which in turn increases the costing.
- Platinum is expensive.

DISCUSSION QUESTIONS

1. How does the SHE facilitate the determination of standard electrode potentials for other half-reactions?
2. Can alternative reference electrodes replace the SHE in certain electrochemical measurements? What are the implications?
3. How does the standard hydrogen electrode potential help in predicting the spontaneity of redox reactions?
4. Discuss the practical applications of the SHE in various fields such as industry, environmental monitoring, and electrochemical research.
5. How does the SHE relate to the concept of the electrochemical series?